2nd OpenFOAM HPC Challenge (OHC-2): Software Track Summary

JULY 1-2, 2026, Guimaraes

Mini-symposium to be held in the auspices of the
21st OpenFOAM Workshop, Guimarães, Portugal

Organized by the OpenFOAM HPC Technical Committee (TC)

Presented by:
Sergey Lesnik, Gregor Olenik, Mark Wasserman

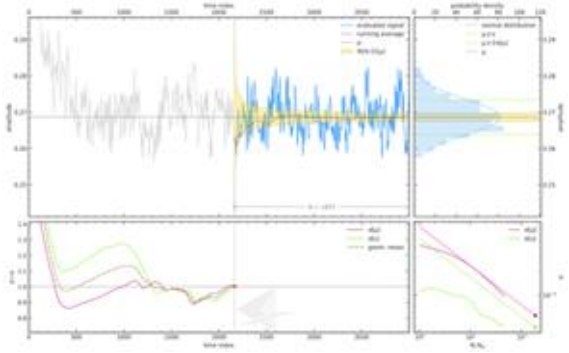
Rules (Software/Open Track)

- **Objective: Showcase code optimizations that improve performance (time, energy), whilst retaining the basic level of accuracy as demonstrated with standard OpenFOAM**
- **Hardware: unconstrained**
- **Software: unconstrained, OpenFOAM-based code**
- **Case setup: occDrivAerStaticMesh, with fixed mesh and physical modelling (temporal and spatial discretization, turbulence model)**
  - Steady-state, incompressible flow
  - RANS with kw-SST turbulence model
  - Pre-defined fvSchemes (changes allowed but have to be reported)
- **Proposed topics for investigation: accelerators, pre-/post-processing (I/O), mixed-precision, linear solvers, renumbering/decomposition**

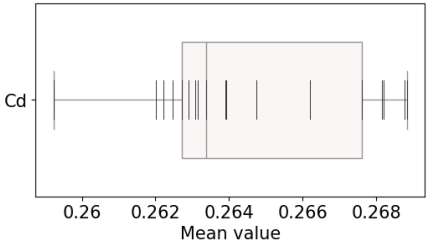
Validation via Force Coefficients

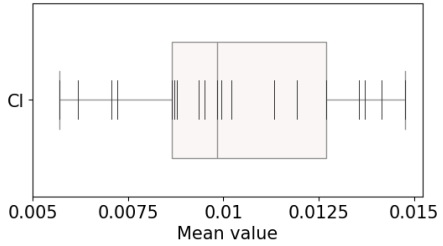
OF Reference Values (fine)
Cd | 2.62E-01
Cl | 7.87E-02
Cs | 1.16E-02

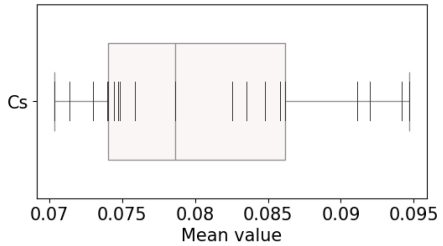
- Extracted profiles at run time
- Participants of the software track were required to extract force convergence and analyze using meanCalc
- Results were reported via submission sheet and time series
- Data analysis team double checked and cross compared values
- Reported data with requirements
- Check validity of 2*sigma(mu) < 0.0015



Total timesteps [nread]	4000	4000	4000
Averaging start timestep [nskip]	1723	1723	1723
Averaging samples [nused]	2277	2277	2277
Mean value μ	0.2627	0.0740	0.0149
Error Mean Value $\sigma(\mu)$	0.00065	0.00165	0.00166
95% conf. int. on mean $2*\sigma(\mu)$	0.0013	0.0033	0.0033
Standard deviation s	0.0052	0.0180	0.0088
Error Standard Deviation $\sigma(s)$	0.0005	0.0012	0.0012







Contributors

- **58 data points contributed by 5 contributors:**
  - Engys
  - CINECA
  - Wikki GmbH
  - TUM
  - HLRS

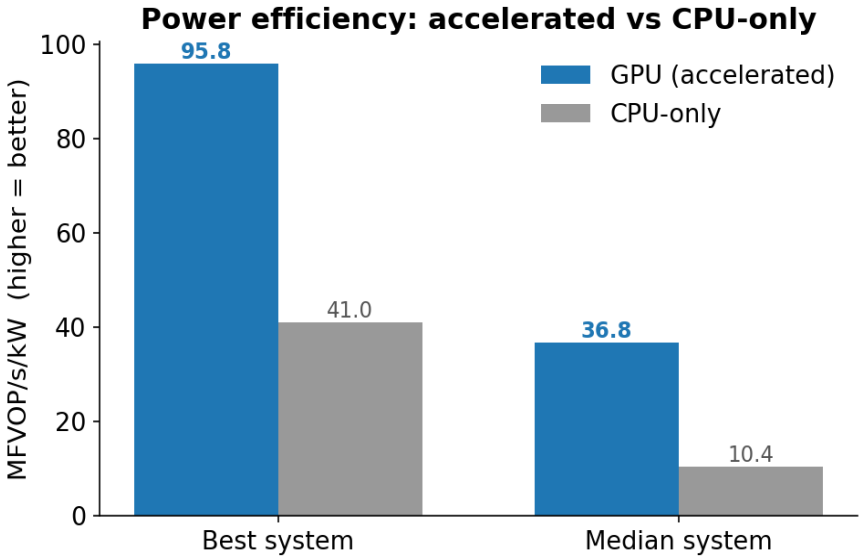
Thank you!

Software Track in Numbers

	Software Track (vs. OHC1)	Hardware Track (vs. OHC1)
Submitted Points	62 (+0%)	86 (−50%)
Min. energy per iteration [Ws]	678 (−63%)	2061 (−21%)
Min. time per iteration [s]	0.07 (−35%)	0.27 (−27%)
Max. performance [MfvOps/kW]	95.8 (+175%)	41.0 (+41%)

Accelerated OpenFOAM Redefines Energy Efficiency

- **Best GPU run: 95.8 MFVOPs/kW (H200, full-GPU port)**
  - ≈ 2.3× the best CPU (41.0 MFVOPs/kW)
- **Median GPU ~3.5× more power-efficient than median CPU**
- **All top-10 efficiency slots are GPU-accelerated**
- **Energy-to-solution is where GPUs pull ahead, while large-scale time-to-solution is still dominated by CPUs**



Metrics

- **Software Track**
  - Single- and Multi-node Scalability
  - Time-to-solution, excl. Pre-processing (decomposition, renumbering) and initialization (potentialFoam)
  - Energy-to-solution (energy-per-iteration)
  - FVOPS (Finite-volume operations per second)
  - **FVOPS / W = FVOP/(s*W) = FVOP/J = Finite-volume operations per J of energy**
  - Accuracy w.r.t reference solution obtained with standard OpenFOAM (hardware track)
    - Mean and variance of aerodynamic coefficients

Power measurements

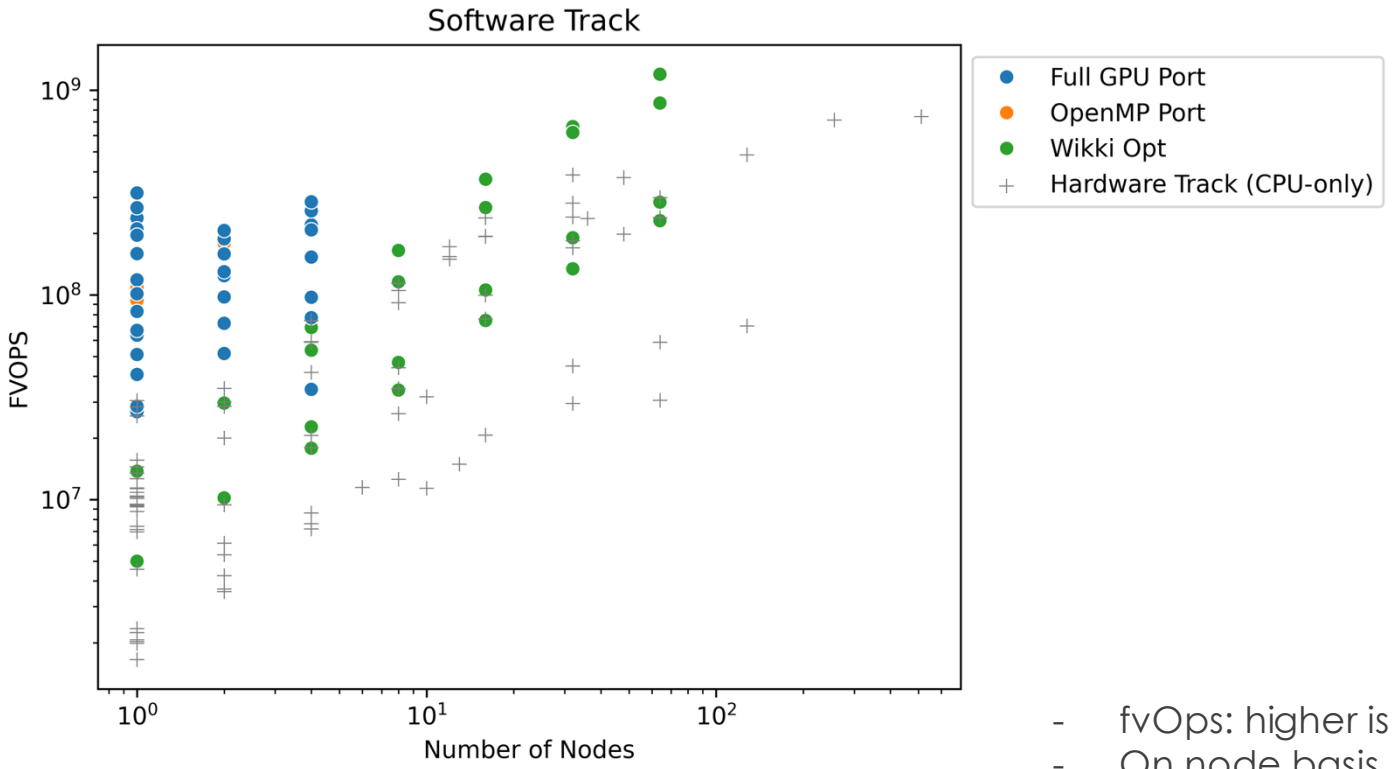
Metric	1×GH200 (65M/GPU)	2×GH200 (32.5M/GPU)	2×H100 (32.5M/GPU)	8xH100 (30M/GPU)
TDP	700 W	700 W/GPU	700 W/GPU	700 W/GPU
GPU power (avg)	481.9 W	444.6 W/GPU	422 W/GPU	410 W/GPU
Percentage of TDP	68.8%	63.5%	60%	58.5%

*** Avg. CPU Usage: 1.4%**

Metric	Grace 72c (65M)
TDP	250 W
CPU power (avg)	245 W
Percentage of TDP	98%

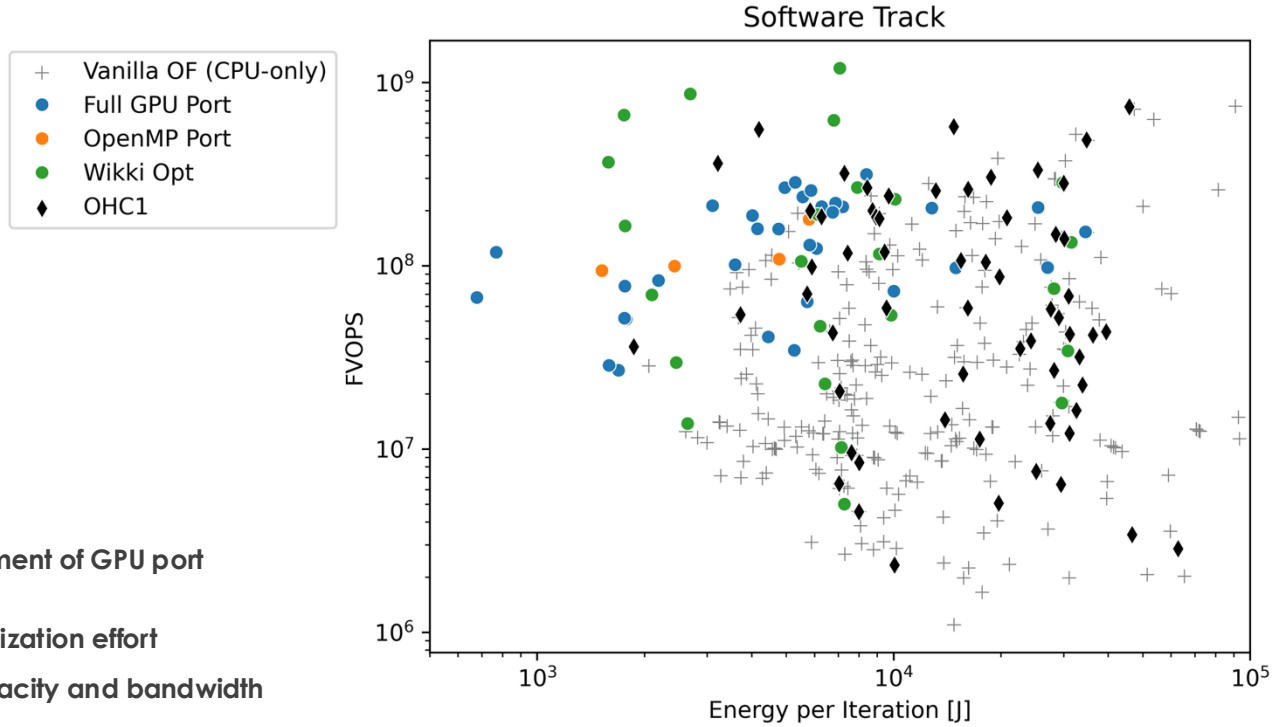
*** Avg. CPU Usage: 99%**

Breakdown of Contributions (Statistics)



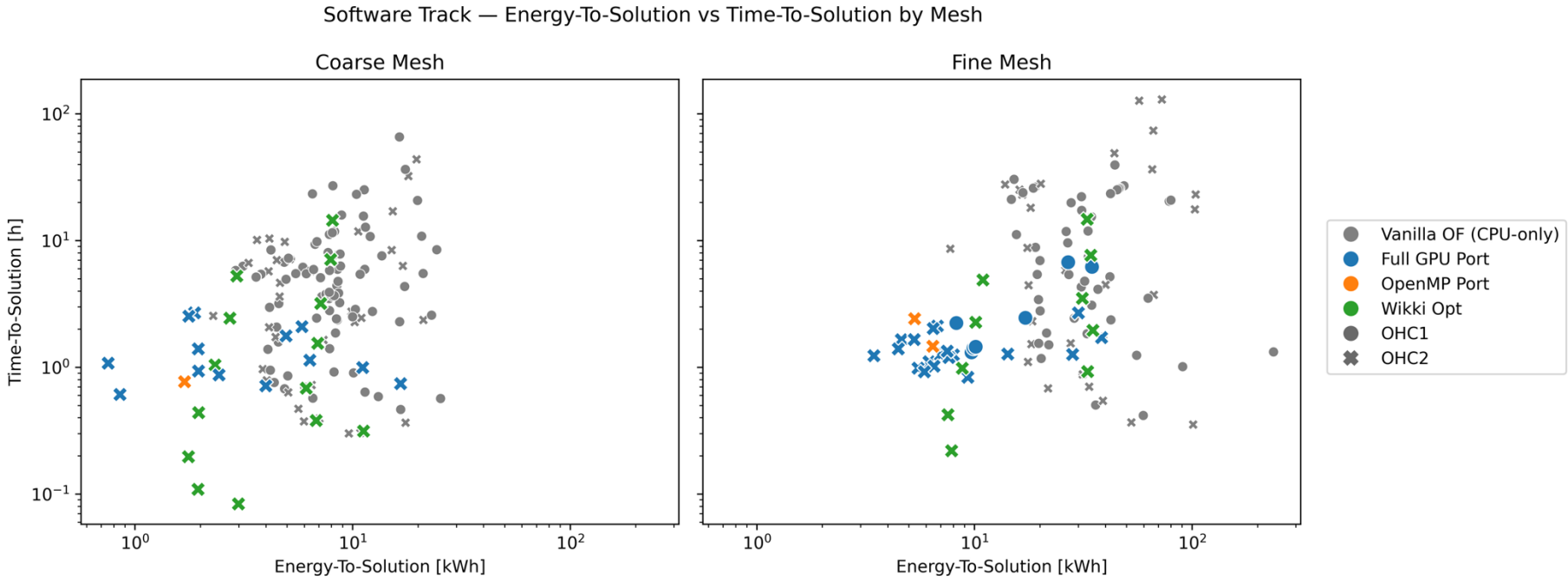
- fvOps: higher is better
- On node basis

Breakdown of Contributions by Category



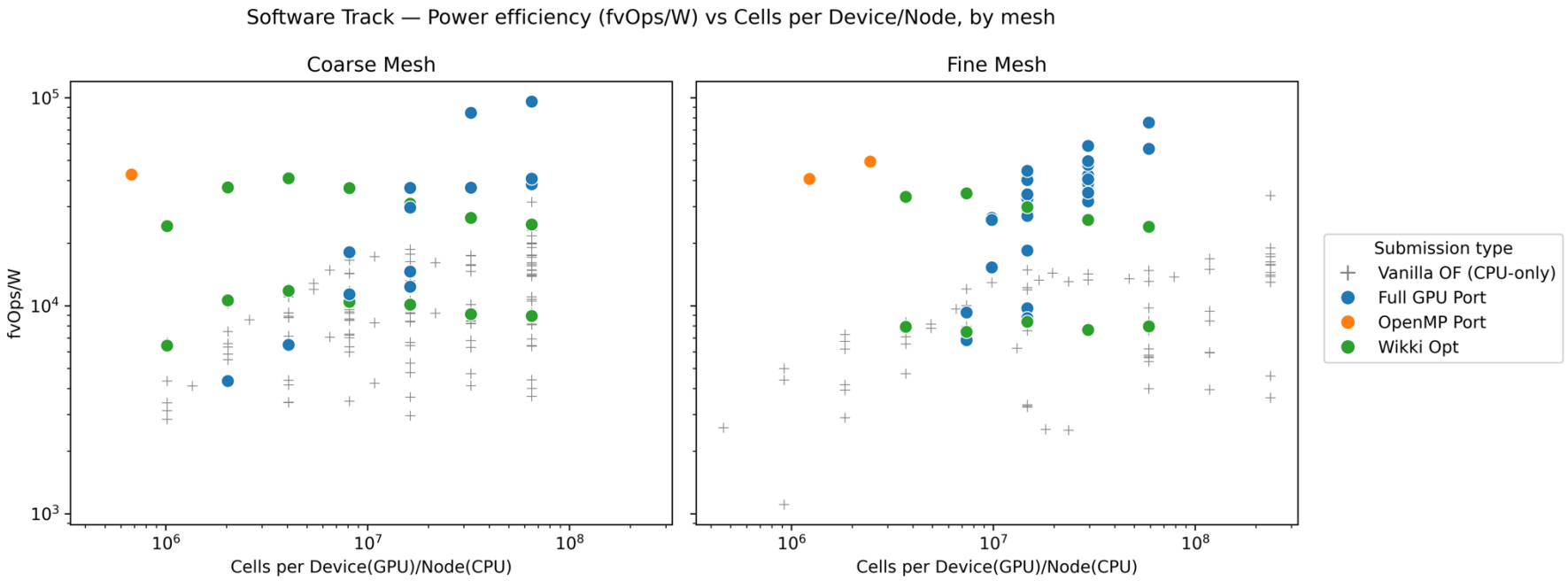
- **Significant enhancement of GPU port performance**
- **Major software optimization effort**
- **Increase in HBM capacity and bandwidth**

Comparison of Optimization Strategies



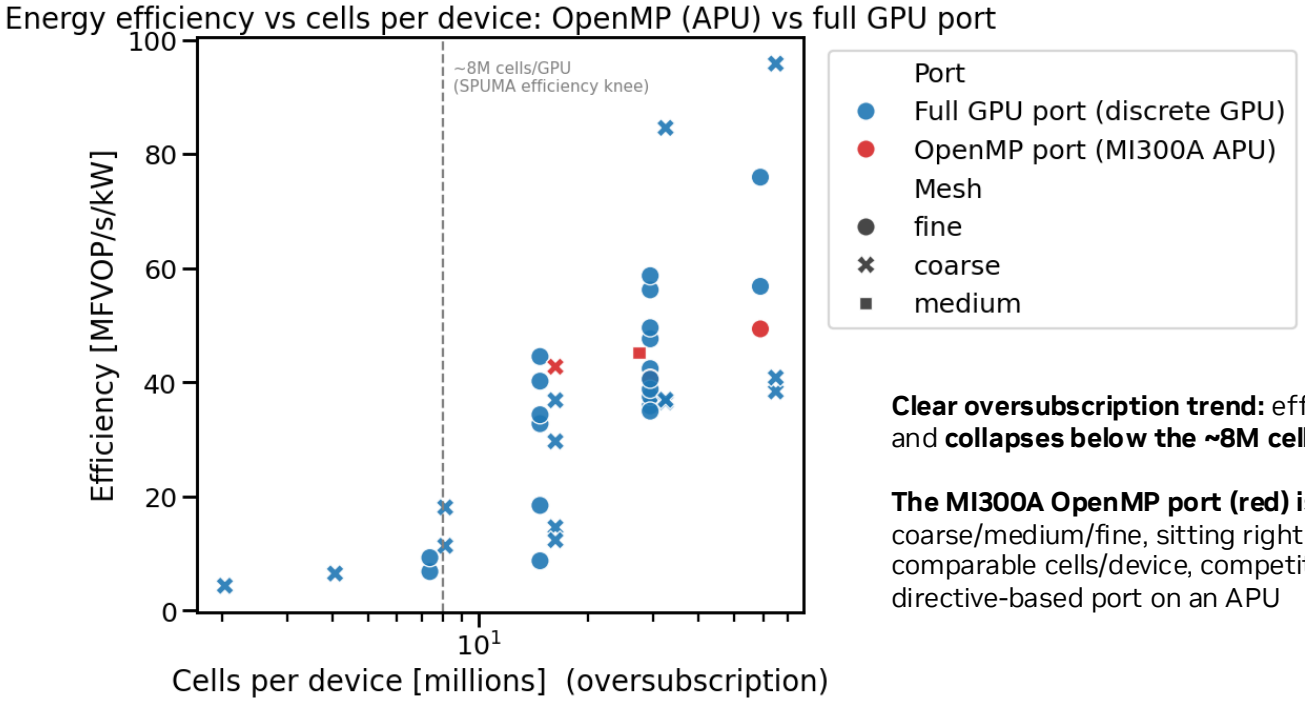
No GPU ports on coarse mesh in OHC1

Accelerators: Cells per Device Metric



- Different optimal settings for CPU and GPU nodes
- Optimal value largely determined by fitting the problem into fast memory tiers (LLC / HBM), and ensuring enough work for accelerator

Accelerators: Cells per Device Metric

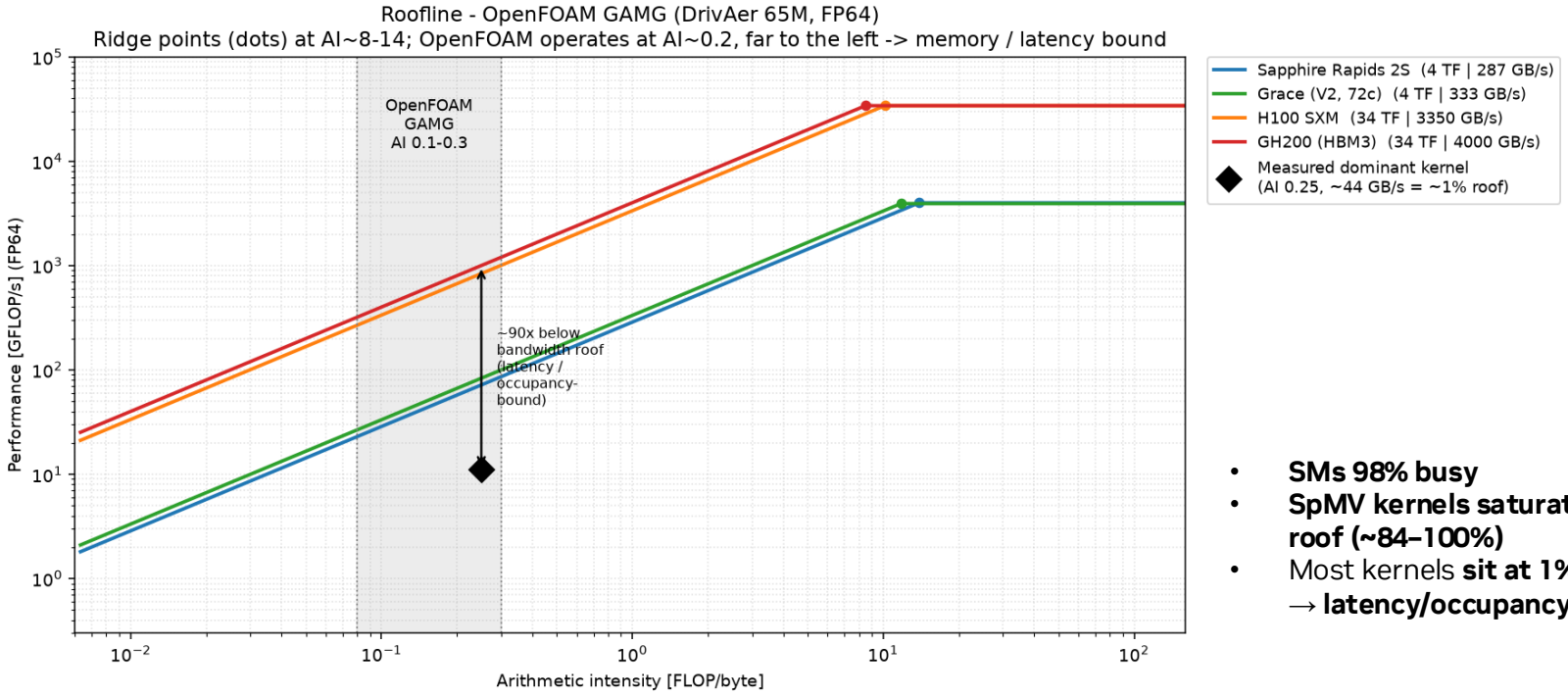


Clear oversubscription trend: efficiency climbs with cells/device and **collapses below the ~8M cells/GPU knee**

The MI300A OpenMP port (red) is stable: ~42–49 MFVOP/s/kW across coarse/medium/fine, sitting right among the discrete-GPU ports at comparable cells/device, competitive despite being a minimal-effort directive-based port on an APU

*https://arxiv.org/pdf/2512.22215v1

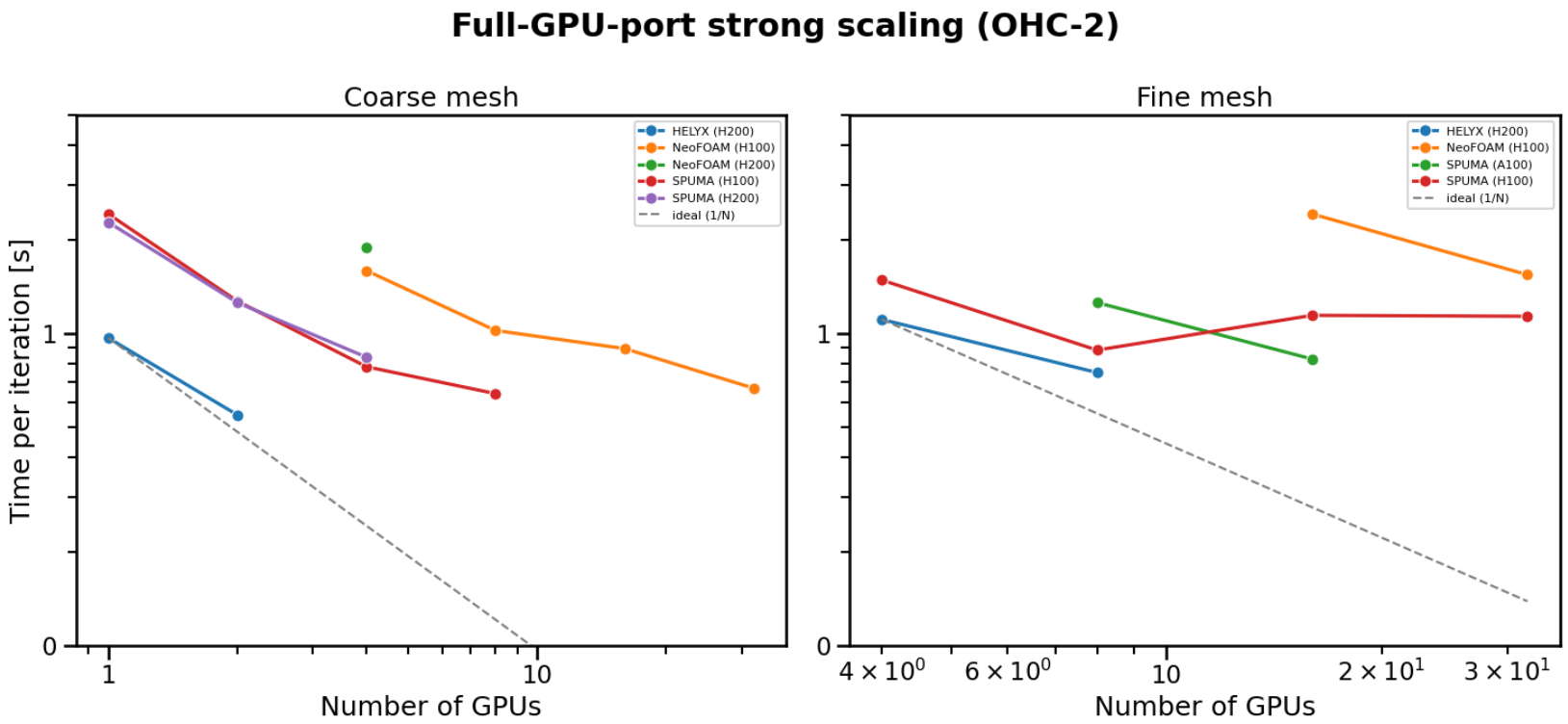
GPU Ports – Roofline analysis



- **SMs 98% busy**
- **SpMV kernels saturate the HBM3 roof (~84–100%)**
- Most kernels **sit at 1% of the roof → latency/occupancy-bound**

Roofs: STREAM-triad BW for SPR/Grace (measured); HBM3 spec for H100 (3.35 TB/s) & GH200 (4.0 TB/s); spec FP64 peak.
Operating point: per-kernel Nsight Compute on 1xGH200 (SPUMA GAMG). Dominant kernel sits at AI~0.25 but only ~1% of the HBM roof -> latency/occupancy-bound, not bandwidth-saturated. (ncu isolated replay flushes L2 & locks clocks -> BW is a lower bound.)

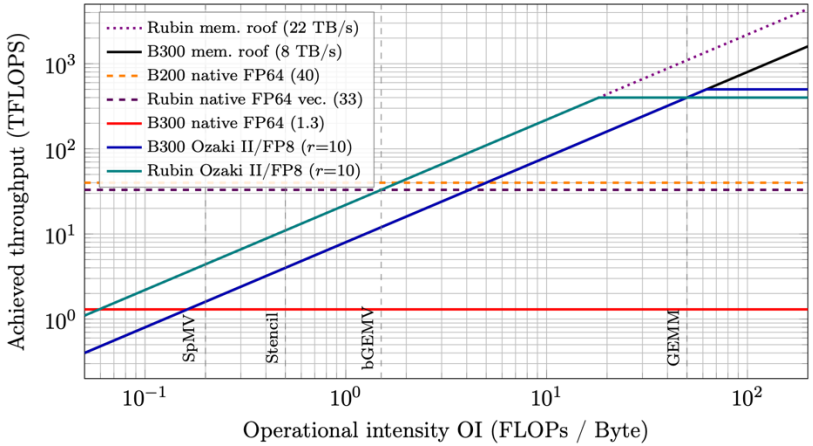
GPU Port Multi-node Scalability



Mixed-Precision

- **Double precision (FP64) remains foundational for scientific computing, simulation, and physics-based workloads.**
- **Current-generation anchor** NVIDIA Ampere A100 introduced third-generation Tensor Cores with FP64 support via DMMA instructions.
- **A100 performance** Up to 19.5 FP64 TFLOPS and up to 2.5× FP64 acceleration versus V100 for HPC matrix math and iterative solvers.
- **Next generation** NVIDIA Blackwell GPUs deliver 30% more FP64 and FP32 FMA performance than Hopper, extending simulation-heavy HPC workloads.
- **Positioning takeaway** NVIDIA sustains broad FP64 support — A100 marks the Tensor Core FP64 inflection point; Blackwell advances overall scientific-computing throughput.

Config	DP [s/iter]	SPDP [s/iter]	Speedup
A100 / 8 GPU	1.82	1.26	**1.45×**
A100 / 16 GPU	1.07	0.83	**1.30×**
H100 / 8 GPU	1.12	0.89	**1.27×**



https://arxiv.org/pdf/2606.06510v1

ISC 26 Student Cluster Competition

- **Virtual competition and on-site Hackathon focusing on optimizing CPU and GPU-ports of OpenFOAM**
- **Improved performance by 30% over baseline versions**
- **Under tight power budget of 6 kW**



- **Mesh decomposition / graph partitioning (biggest lever)**
  - kahip (03-ETH) — built ThirdParty kahip; a clean controlled comparison showed ~27% speedup vs hierarchical on identical 8×H100 (971→708 s)
  - scotch (01-CHPC) as an alternative to geometric hierarchical
- **GPU kernel-launch tuning**
  - -nThreadsPerBlock block-size tuning: 128 (09-THU) and 512 (10-NTU) — a SPUMA-specific CUDA knob that gave measurable gains on top of binding
- **Rank binding / topology-aware placement**
  - NUMA + UCX binding, MPI rankfile binding (01-CHPC), and topology-aware GPU/CPU binding + NCCL tuning (08-SYSU); per-rank GPU pinning generally
- **Memory pool & I/O**
  - fixedSizeMemoryPool poolSize tuning and hostCollated file handling to cut I/O overhead (06-NUS — the fastest run, 483 s).
- **Power management (for the 6 kW budget)**
  - GPU power-capping: 05-NTHU swept H200 TDP with nvidia-smi -pl; 01-CHPC ran a 400 W cap; 02-EPCC power-capped its CPU nodes — trading a little speed for staying within budget (and maximizing perf/W).
- **Hardware/scale selection & build verification**

Conclusions and Future Work

- **Significant increase in percentage of software track contributions**
- **OpenMP and GPU ports have made a huge leap over the past year**
- **Wikki optimizations demonstrate that there is still room for improvement of CPU-only OpenFOAM**
- **Software optimizations can outperform several generation-jumps in hardware**
- **OccStaticDrivAer benchmark analyzed in many publications over the past year**
- **Efficiency (MFVOP/s/kW) is the emerging differentiator, beyond raw speed**
  - Need to standardize energy reporting
- **GPU ports still have much room for improvement**
  - Kernel launch, fusion and multi-node scalability
  - Happy to support anyone working on GPU porting